Arturo Arellano Arias

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Education

Ph.D. degree in Mathematics — *McGill University*
GPA: 4.0 — *September 2021 - present*
Thesis project: Shape theorems for periodic branching diffusions
Advisors: Jessica Lin and Louigi Addario-Berry

Master's degree in Applied Mathematics — *CIMAT, Mexico*
GPA: 3.8 — *August 2019 - August 2021*
Thesis: Percolación, el modelo de Ising y SLE
Advisor: Juan Carlos Pardo

Bachelor's degree in Mathematics — *Universidad de Guanajuato, Mexico*
GPA: 3.5 — *August 2014 - August 2019*
Thesis:Una mirada al Teorema Ergódico Subaditivo
Advisor: Renato Iturriaga

Awards and fellowships

FRQNT Doctoral Research Scholarship — *May 2023 - December 2026*
Fonds de recherche du Québec – Nature et technologies

MITACS Globalink Research Award — *April 2023*
MITACS

CONACYT National Graduate Scholarship — *August 2019 - August 2021*
Consejo Nacional de Ciencia y Tecnología, Mexico

CIMAT Undergraduate Scholarship for Excellence — *August 2014 - August 2019*
CIMAT

Gold medal Mexican Mathematics Olympiad — *November 2013*
Sociedad Matemática Mexicana

Publications and preprints

A. A. A. **"A shape theorem for periodic branching diffusion"** — *August 2026*
arXiv:2608.17031 • *Submitted to ALEA Latin American Journal of Probability and Mathematical Statistics*

J. Amorim, A. A. A, M. Jara. **"Scaling limits of Smoluchowski particles"** — *March 2026*
arXiv:2603.25868 • *Submitted to Electronic Journal of Probability*

L. Addario-Berry, A. A. A, J. Lin. **"A shape theorem for BBM in a periodic environment"** — *July 2025*
arXiv:2507.10515 • *Submitted to Annales de l'Institut Henri Poincaré*

A. A. A., H. Chang-Lara. **"A discrete scheme for a transmission problem combining a first-order and a second-order operator"** — *In preparation*

Teaching experience

Instructor, MATH 203: Principles of Statistics *Summer 2026*
McGill University *Montreal, Canada*
Sole instructor of an introductory undergraduate statistics course with 92 students from a wide range of programs. Topics included descriptive statistics, common distributions, and confidence intervals and hypothesis tests for means, variances and proportions. Designed the lectures, assignments, exams and grading scheme, and managed course logistics, including the course website, exam scheduling and accommodations for students with disabilities.

Teaching Assistant, MATH 447: Introduction to Stochastic Processes *Winter 2026*
McGill University *Montreal, Canada*
Sole teaching assistant for an upper-level undergraduate course in stochastic processes with 91 students. Topics included conditional expectation, generating functions, branching processes, random walks, Markov chains and the ergodic theorem, birth-death processes and queueing theory. Designed and led weekly 1.5-hour tutorials, including a dedicated final exam review, and held weekly office hours.

Teaching Assistant, MATH 323: Probability *Fall 2024*
McGill University *Montreal, Canada*
One of three teaching assistants for an undergraduate probability course with 494 students from a wide range of programs. Topics included conditional probability and Bayes' theorem, combinatorial probability, discrete and continuous random variables and distributions, independence, probability inequalities, the law of large numbers and the central limit theorem. Designed and led weekly tutorials, held weekly office hours, and worked with the instructor and the other teaching assistants to coordinate tutorial content and grading.

Teaching Assistant, MATH 315: Ordinary Differential Equations *Fall 2022*
McGill University *Montreal, Canada*
Sole teaching assistant for an undergraduate course in ordinary differential equations with 100 students. Topics included first-order equations and elementary numerical methods, linear differential equations, Laplace transforms and series solutions. Led a review session for the midterm exam.

Teaching Assistant, Elements of Geometry and Calculus I, II and III *2018 – 2021*
Universidad de Guanajuato *Guanajuato, Mexico*
Sole teaching assistant for four first- and second-year courses for mathematics majors: Elements of Geometry (Euclidean geometry), Calculus I and II (single-variable differential and integral calculus) and Calculus III (multivariable calculus). Taught in Spanish.

Mentoring

Directed Reading Program *Winter 2023, Winter 2024*
Graduate Mentor *McGill University*
Mentored two undergraduate students in semester-long reading projects on Monte Carlo Simulations, and Stochastic processes, through weekly one-on-one meetings.
Guided each student in writing an expository paper and preparing a final presentation.

Coach Mexican Mathematics Olympiads *2013 - 2021*
Coach and Member of the State Organizing Committee *Guanajuato and Campeche, Mexico*
Organized the state olympiad at every stage, from the initial school-level rounds to the final selection of the six students representing the state at the national competition.
Coached students in olympiad problem solving through regular training sessions.

Talks and Posters

A shape theorem for the convex hull of d-dimensional branching Brownian motion in periodic environments

- BIRS Workshop on Reaction-Diffusion, Branching Processes, and Biology, Banff *May 2025*

- SIAM Student Chapter, Georgia Tech *April 2025*
- CMS Winter Meeting *December 2024*

Branching Brownian motion in a periodic environment (Poster)

- Conference on New Developments in Probability, CRM, Montreal *September 2024*

A shape theorem for multidimensional branching Brownian motion

- Students' Probability Seminar, IMPA, Rio de Janeiro *May 2024*

A numerical method for the solution of a transmission problem

- Students' Seminar in Applied Mathematics, McGill University *May 2023*

Un algoritmo para aproximar la mejor estrategia en un juego estocástico

- Junior Seminar, Universidad de Guanajuato *November 2022*

Technical skills

Programming Languages/Tools Python, R, GitHub, LaTeX

Languages

English Fluent
Spanish Native